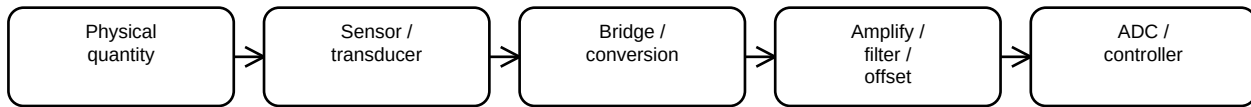


- Define reliability and explain the exponential law of reliability.
- Discuss failure causes during infant mortality, useful life and wear-out.
- Explain System Performance Matrix and Design Matrix use.
- Analyze the supplied inverting-amplifier problem: $R_1=10\text{ k}\Omega$, $R_f=20\text{ k}\Omega$, $V_i=3\text{ V}$, $2\text{ k}\Omega$ load and $I_Q=0.5\text{ mA}$; calculate the requested current/power quantities using the problem's assumptions.
- Explain TTL-to-CMOS and CMOS-to-TTL interfacing with suitable circuits.
- Explain VOS, I_B , IOS and CMRR and their influence on measurement.

UNIT 2 — ANALOG HARDWARE DESIGN

1. Signal Conditioning



The course repeatedly uses signal conditioning because a sensor output may not directly match the required voltage/current range, accuracy or drive capability of the next stage.

Design checklist: sensor range → signal type → required output range → gain → offset → filtering → reference/excitation → protection → verification.

2. V→I Converter — 4–20 mA

Supplied design problem: deliver 4–20 mA to a floating load for 0–4 V input.

V _{in}	Required I _{out}
0 V	4 mA
1 V	8 mA
2 V	12 mA
3 V	16 mA
4 V	20 mA

The output span is 16 mA for a 4 V input span, so the required slope is 4 mA/V. The 4 mA lower endpoint is an intentional offset; it is not a zero-current output.

Interactive: V_{in} slider → current through floating load → I_{out} graph. Add separate gain/span and offset controls and make the learner calibrate 4 mA and 20 mA endpoints.

Exam design: derive transfer relation, select component values, verify lower/upper endpoints and test an intermediate point.

3. I→V Converter — 4–20 mA to 0–4 V

The design task is to map the 4–20 mA input range to 0–4 V. The key reasoning is to remove the 4 mA baseline and scale the 16 mA span to 4 V.

Interactive: current slider → raw current-to-voltage conversion → offset subtraction → gain → final 0–4 V. Show every stage numerically.

4. Gain and Offset / Zero-and-Span

The supplied material describes a zero-and-span circuit for producing a convenient parameter-to-voltage relationship when the transducer output does not match the ADC requirement. A linear mapping can be represented as $y=mx+c$.

Why? A sensor may be linear but have the wrong slope, starting point, or both. Gain and offset conditioning maps it into a useful measurement range.

Interactive: define sensor minimum/maximum and target minimum/maximum. Compute slope and intercept, then let the learner choose an implementation.

5. RTD Signal Conditioning

The supplied RTD material writes resistance as $R+\Delta R=R(1+\delta)$, where $\delta=\Delta R/R$. For the PT100 calculations it uses $R(0^\circ\text{C})=100\ \Omega$ and $\alpha=0.00392/^\circ\text{C}$, giving $R(T)=100(1+0.00392T)\ \Omega$.

T	R(T) from supplied example
25°C	109.8 Ω
100°C	139.2 Ω
-15°C	94.12 Ω

The material explains bridge conversion of ΔR to ΔV and then instrumentation-amplifier amplification of the bridge difference.

Self-heating: the supplied design problem limits PT100 power dissipation below 0.2 mW and selects approximately 1 mA for safety, leading to $R1\approx 15\ \text{k}\Omega$ with a 15 V reference.

Interactive: temperature $\rightarrow$ resistance $\rightarrow$ bridge imbalance $\rightarrow$ IA output. Add lead resistance and self-heating controls.

6. RTD Design and Calibration Example

Supplied problem: PT100 in a bridge driven by 15 V; compute R1 and IA gain for 0.1 V/°C sensitivity; estimate output at 100°C and equivalent error using the approximate equation.

The supplied calculation uses $\delta = \alpha \Delta T = 0.00392$ for 1°C and chooses gain so the desired output change is 0.1 V/°C.

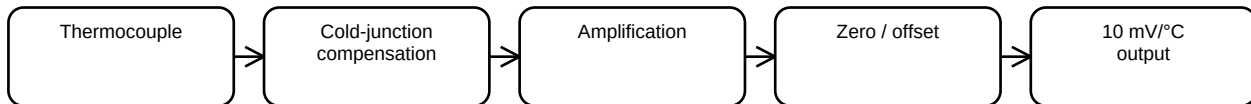
For the 100°C comparison, the supplied material gives approximately 9.974 V from the more complete relation versus 10.0 V from the approximation, an error of 0.026 V. Under the stated 0.1 V/°C scaling this corresponds to about 0.26°C.

Calibration sequence from the material: PT100 at 0°C → adjust R2 for zero IA output; PT100 at 100°C → adjust Vref for Vo=10 V.

Interactive: ask for each calculation before showing the supplied solution. Include a calibration mode with two reference temperatures.

7. Thermocouple Signal Conditioning

Supplied design problem: K-type thermocouple conditioner with 10 mV/°C output sensitivity, 0 mV at 0°C, and cold-junction compensation.



Why cold-junction compensation? A thermocouple responds to the temperature difference associated with its junctions; compensation is therefore needed when the desired output is referenced to a defined temperature scale.

Interactive: change measurement-junction and reference-junction temperature and observe the compensated output.

8. Load Cell

The supplied material describes a strain-gauge load cell as a force/weight sensor. A spring element creates high-strain regions where strain gauges are attached. Deflection changes resistance and produces a ratiometric bridge output.

The output is specified in mV/V. A stable excitation voltage is applied to opposite Wheatstone-bridge corners and the differential signal is measured across the other corners.

Important specifications to teach: rated load/range, sensitivity, excitation, accuracy, nonlinearity, hysteresis, repeatability, creep, zero balance and temperature effects. Do not invent course-specific numerical values where none are supplied.

The supplied material also discusses sense wires for monitoring actual excitation voltage, especially with long cable runs.

Interactive: load → strain → resistance change → bridge output → instrumentation amplifier → ADC.

9. Solid-State Relay / Optically Coupled SSR

An SSR uses semiconductor switching rather than a mechanical contact. Optical coupling transfers the control signal across an isolation barrier, separating the control electronics from the load-side circuit.

Why? Isolation helps protect low-voltage control electronics from the load-side electrical environment and allows the two sides to have different electrical domains.

Interactive: controller → optical isolation → semiconductor switch → load. Clearly shade the isolation boundary and show the state transition when the input toggles.

10. Instrumentation Amplifier

The supplied material uses a three-op-amp instrumentation amplifier to amplify V_1-V_2 while rejecting common-mode voltage. It also presents single-chip INA devices and the relationship $V_o=G(V_{IN+}-V_{IN-})+V_{ref}$.

Derive the classic three-op-amp IA in stages: input amplifier pair $\rightarrow$ differential amplifier $\rightarrow$ total gain. Explain resistor matching because mismatch degrades common-mode rejection.

Interactive: differential input, common-mode input and gain-resistor sliders. Show differential output, common-mode error and saturation.

11. INA118 / INA128 / INA333 Examples

Parameter	INA118	INA128	INA333
Offset	50 μ V max	50 μ V max	25 μ V
Drift	0.5 μ V/ $^{\circ}$ C max	0.5 μ V/ $^{\circ}$ C max	0.1 μ V/ $^{\circ}$ C
Input bias	5 nA max	5 nA max	200 pA max
CMR figure	110 dB min	120 dB min	100 dB min
Supply	± 1.35 to ± 18 V	± 2.25 to ± 18 V	1.8 to 5.5 V
Quiescent current	350 μ A	700 μ A	50 μ A

These are examples from the supplied material. Use them in an interactive specification-selection exercise: the learner enters constraints and sees which devices satisfy those constraints. Do not turn the table into a universal recommendation.

12. ON–OFF Controller

The supplied controller material defines set point (SP), process variable (PV), error $E = SP - PV$ and controller output. ON–OFF control drives the actuator fully ON or OFF and uses hysteresis/dead band to prevent excessive chattering.

Why hysteresis? If switching thresholds are identical, noise and small process movement around the set point can produce rapid actuator switching. Hysteresis creates distinct upper and lower thresholds.

The material gives the Schmitt-trigger relation $\Delta E \approx \beta V_{osat}$ with $\beta = R2/(R1+R2)$ for the shown circuit concept.

Interactive: move PV through the hysteresis band and watch the relay state. Add a noise slider to demonstrate why the band exists.

13. Supplied ON–OFF Temperature Controller Design

Problem values: 0–400°C span; signal conditioner 10 mV/°C; adjustable set point; 3-digit DPM, 0–1 V input; hysteresis 5% of span; 12 V, 60 Ω relay actuator with 2 CO, 10 A AC contacts.

The supplied solution maps temperature to 0–4 V, uses a 4.096 V reference adjusted to 4 V, attenuates the process signal for the display, and creates a total hysteresis of 0.2 V using ± 0.1 V thresholds.

For the relay driver the supplied solution calculates about 200 mA coil current, about 1 mA base current for the selected BC337 stage, and an approximately 10 k Ω base resistor. It calculates a temperature-adjusted leakage value and selects about 15 k Ω for the second base resistor.

The supplied solution uses 1N4007 protection: one around the base-emitter network for negative saturation protection and one across the relay coil for voltage-reversal spikes.

Interactive: calculate every stage; clicking a component opens “Why this value?” and “What if this value changes?”.

14. Proportional Controller

The supplied material states that proportional control provides a linear relationship between controller output and error over the proportional band.

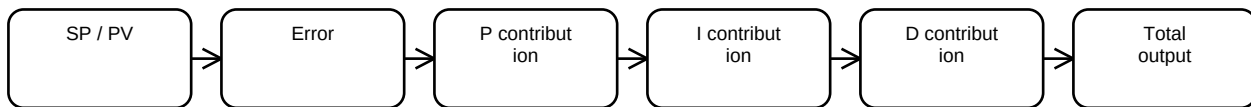
It gives $P = K_p \cdot e_p + P_o$ for the stated action convention, with K_p as proportional gain and P_o as controller output at zero error.

Limitation: the supplied material states that proportional control alone cannot generally eliminate steady-state error; a load change can create a permanent offset.

Interactive: error-output graph; change K_p and P_o . Show proportional band and offset.

15. PI and PID

The supplied material states that integral action is added to proportional control to remove steady-state error, while integral-only action can respond sluggishly. PID adds derivative action for improved transient behavior.



Interactive controller: show P, I, D and total traces separately. Let the learner change one parameter at a time and observe the response.

Supplied PI design example: 0–400°C span, 10 mV/°C signal conditioner, 20% proportional band, integrator RC time of 2.2 minutes, 0–10 V output and adjustable set point.

Supplied PID design material includes the 0–400°C span, 0–4 V signal-conditioning range and proportional-gain calculations. Preserve the source's sign/action convention when implementing the example.

16. PID Tuning Lab

The website should teach tuning as an experiment rather than a memorization table: choose K_p , K_i and K_d → run the model → observe rise/settling/overshoot/steady-state error → change one parameter → rerun.

Use separate visual explanations for present error (P), accumulated error (I) and rate-of-change of error (D). Include a noise experiment to show why derivative action can be sensitive to rapid signal changes.

The simulation is an educational model. It should not be presented as a universal tuning result for every physical plant.

17. Unit 2 Practice Bank

- Design 0–4 V to 4–20 mA V–I converter for a floating load.
- Design the supplied pneumatic-tank ON–OFF controller: sensor 0–2 V for 0–10 kg/cm², SP 6 kg/cm², hysteresis ± 0.2 kg/cm² and 0/12 V output.
- Design K-type thermocouple conditioner for 10 mV/°C and 0 mV at 0°C with cold-junction compensation.
- Derive instrumentation-amplifier gain and discuss characteristics/applications.
- Discuss important load-cell specifications.
- Explain why relay coils require voltage-spike protection.
- Compare ON–OFF, P, PI and PID in terms of error and response.

UNIT 3 — DIGITAL HARDWARE DESIGN

1. ADC Specifications

The syllabus requires interpretation of important ADC/DAC specifications. The supplied material includes a 10-bit SAR ADC with 4.094 V reference as a numerical example.

Specification	Learning target
Resolution	Smallest represented voltage step; linked to number of bits and reference range.
Reference voltage	Sets the conversion scale/range.
Quantization	Maps continuous analog values to discrete digital codes.
Offset error	Shift of the conversion characteristic.
Gain error	Scale/slope deviation.
Linearity	Closeness of actual transfer to ideal transfer.
Sampling rate	Acquisition rate; must be considered with input bandwidth and aliasing.

Interactive: bit-depth, V_{ref} and V_{in} sliders → code, step size and quantization visualization.

Course numerical: 10-bit SAR ADC, $V_{ref}=4.094$ V. Ask the student to calculate resolution using the course convention and make the denominator assumption explicit.

2. DAC

Teach DAC as conversion from digital code to analog output using a reference. Important dimensions include resolution, reference accuracy, offset/gain error, linearity and settling behavior.

Why reference? The reference establishes the scale of the analog output. Reference error or drift therefore affects conversion accuracy.

Interactive: binary switches + reference slider + analog output staircase.

3. Anti-Aliasing Filter

The supplied material states that sampling frequency should be greater than twice the highest frequency component of interest. It explains that components above $f_s/2$ can fold back into the band of interest as aliases.

Supplied example: a 5 MHz sine wave sampled at 6 MS/s can appear as a 1 MHz signal because the Nyquist frequency is 3 MHz.

Why filter before ADC? Once unwanted high-frequency content has folded into the sampled band, the digital system cannot reliably tell the alias from a genuine in-band signal. Filtering must therefore occur before sampling.

Interactive frequency-domain lab: signal frequency, sampling frequency and filter cutoff sliders; show original spectrum, filter response and aliased result.

4. Aliasing Practice

The supplied material gives alias frequency as the absolute difference between the input frequency and the closest integer multiple of the sampling frequency.

Use the supplied example set: $f_s=100$ Hz and input frequencies 25 Hz, 70 Hz, 160 Hz and 510 Hz. Make the learner predict each alias before revealing the calculation.

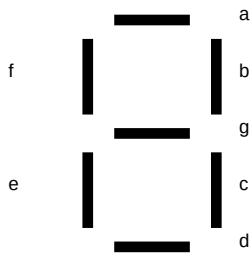
Exam: state Nyquist criterion, define aliasing, explain the anti-aliasing filter and calculate aliased frequencies.

5. Voltage Reference for ADC/DAC

The syllabus requires reference-selection considerations. Turn this into a requirements exercise: reference voltage, tolerance, drift, current capability, temperature behavior and supply constraints.

Use the supplied 4.094 V / 4.096 V examples to show that reference selection is part of the measurement architecture rather than simply 'using the supply'.

6. Seven-Segment Display



The supplied material covers common-anode/common-cathode displays and the 7447 decoder. The 7447 accepts four BCD (8421) input lines and uses open-collector outputs to drive indicator segments.

The supplied material states that the segment outputs have specified sinking capability and withstand capability and that auxiliary inputs support blanking, lamp test and zero-suppression functions.

Why current limiting? Each segment is an LED; current must be controlled by the interface rather than allowed to rise without a defined limit.

Interactive: BCD switches → 7447 behavior → segments. Click any segment to show its label and purpose.

7. 7447 Blanking and Zero Suppression

The supplied material describes RBI/RBO use for leading-zero suppression. It also describes LT and BI/RBO behavior: a LOW on LT turns on all segment outputs when BI/RBO is not forced LOW; BI/RBO can blank all segment outputs.

Interactive: enable/disable LT, RBI and BI/RBO and observe the display. Add a leading-zero suppression mode for multi-digit displays.

The material also gives TTL unit-load values. Include a small calculator converting unit-load factors to currents using the stated 40 μ A HIGH and 1.6 mA LOW definitions.

8. Static Seven-Segment Display

In static display, each digit is continuously driven. It is conceptually simple but uses more dedicated connections as the number of digits grows.

Supplied design problem: design a static display for a product counter. Turn this into a resource-allocation exercise: digits $\rightarrow$ segments $\rightarrow$ drivers $\rightarrow$ logic/MCU connections.

9. Dynamic Seven-Segment Display

Dynamic display rapidly enables one digit at a time while sharing segment lines. The sequence is repeated quickly so the display appears continuously illuminated.

Why? Multiplexing reduces the number of independently controlled segment lines, at the cost of timing, duty-cycle and driver considerations.

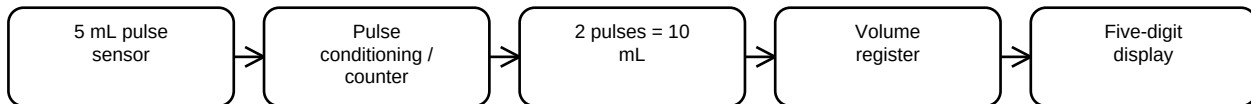
Interactive animation: Digit 1 ON $\rightarrow$ Digit 2 ON $\rightarrow$ Digit 3 ON $\rightarrow$ Digit 4 ON. Add a refresh-rate slider and slow it down to make multiplexing visible.

Teach trade-offs: GPIO count, duty cycle, brightness, current and refresh rate.

10. Petrol / Diesel Dispensing Measurement System

Supplied problem: maximum one-time volume 999.99 L, resolution 10 mL, flow sensor produces one pulse after every 5 mL, static seven-segment display.

Maximum volume is 999,990 mL. At 10 mL resolution, that is 99,999 display increments. Because the sensor produces one pulse every 5 mL, two pulses correspond to one 10 mL display increment.



Interactive: inject pulses; show pulse count, volume in mL/L and display value. Add an overflow state above 999.99 L.

Keep static seven-segment as the main course implementation because that is the stated supplied problem; dynamic display can be a comparison experiment.

11. Microcontroller Selection

Treat MCU selection as requirements matching: I/O count, ADC/DAC, timers, PWM, memory, speed, power, communication interfaces, package/availability, cost and development ecosystem.

Interactive selector: enter system requirements and filter a small set of example devices. Show which constraints each device satisfies instead of declaring one device universally superior.

12. Capacitive Touch — Physical Principle

The supplied material explains capacitive sensing using electric fields and charge-holding conductors. A sensor plate has parasitic capacitance to surrounding conductors. A finger approaching the plate changes the electric-field distribution and therefore the measured capacitance.

For the self-capacitance model in the material, the sensor capacitance changes approximately from C_{base} to $C_{base} + C_{finger}$ when the finger enters the sensing field.

Why? The finger acts as a charge-holding object and changes how the electric field terminates. Electronics detect the resulting electrical change instead of requiring mechanical contact.

The material's sensor-plate design condition is to maximize capacitance between approaching finger and sensor while minimizing capacitance between finger and unrelated ground structures.

Interactive: animate electric-field lines before/after touch and show C_{base} , C_{finger} and total capacitance.

13. ADC-Based RC Touch Sensing

The supplied material describes: (1) set pin as output HIGH to charge capacitors; (2) set pin as ADC input so capacitors discharge; (3) wait a fixed time and read the analog value; (4) decide touched/untouched and restart.

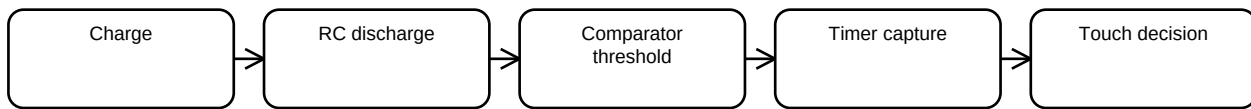
The RC time constant is $\tau = RC$. A change in capacitance therefore changes the charging/discharging curve and the voltage observed after a fixed time.

Interactive: R, C_{base} , C_{finger} and measurement-time sliders → $V(t)$ graph → touch/no-touch decision threshold.

The supplied material notes that very large C_{base} compared with C_{finger} makes the relative touch-induced change small, while very small C_{base} can make other variations/noise more significant.

14. Comparator + Timer Capacitive Sensing

For microcontrollers with an analog comparator and timer capture, the supplied method charges the sensor capacitance, lets it discharge, monitors a threshold with the comparator and captures elapsed time when the threshold is crossed.



Why time? The discharge time depends on RC, so a small capacitance change can be converted into a measurable timing difference.

Interactive: change C and observe threshold-crossing time; add noise and baseline averaging.

15. Self Capacitance

The supplied material defines self-capacitance with respect to a reference conductor and describes parasitic capacitance CP arising from the sensor pad, traces and vias to surrounding conductors such as ground planes and chassis metal.

The material presents self-capacitance as highly sensitive and suitable for single buttons, sliders and proximity sensing, while noting limitations for true multi-touch because the measurement represents total pad change.

16. Mutual Capacitance / PCAP

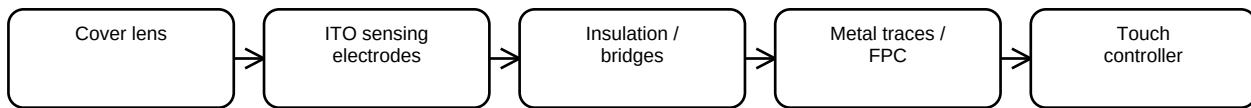
Projected capacitive touchscreens intentionally create mutual capacitance between row and column electrodes near their intersections. The supplied method drives Tx and measures charge on Rx. When a finger is placed between Tx and Rx, mutual capacitance decreases and the received charge decreases.

Why a grid? Measuring individual intersections provides spatial information and supports multiple-touch detection.

Interactive grid: scan Tx rows one at a time, read Rx columns, place virtual fingers and highlight nodes crossing the touch threshold.

17. Projected Capacitive Touchscreen Construction

The supplied material describes an ITO-based projected-capacitive panel patterned into row and column electrodes, with metal traces connecting electrodes to a touch-controller IC, typically through flexible printed circuitry. It also discusses insulating structures/bridges and a protective cover layer.



Interactive cross-section: click each layer to reveal its function; animate a finger approaching an intersection and show the measured change.

18. Four-Wire / Eight-Wire Concepts

The supplied material notes that voltage losses in bus bars and connections can vary with product use, temperature and humidity. In a 4-wire sensor these changes can appear as touch-location error/drift; additional sense lines in an 8-wire arrangement can dynamically reference voltage to correct for the fluctuation.

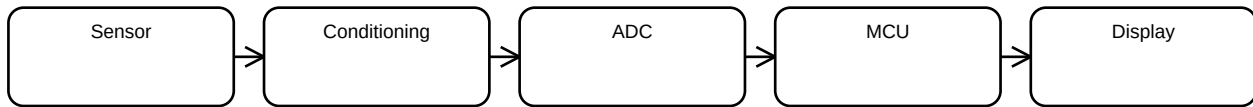
Interactive fault exercise: change connection resistance → observe location error → enable sense lines → observe conceptual correction.

19. Unit 3 Practice Bank

- Design a static seven-segment product counter.
- Explain objectives and operation of dynamic seven-segment display.
- Design the supplied petrol/diesel measurement system with 10 mL resolution and 999.99 L maximum.
- Discuss important ADC/DAC specifications.
- Solve the supplied 10-bit SAR ADC with 4.094 V reference.
- Analyze the supplied gain-100 non-inverting amplifier requirement while considering op-amp parameter errors.
- Explain capacitive touch-screen construction and working.
- Explain self versus mutual capacitance.
- Explain why anti-aliasing filtering is placed before an ADC.

CROSS-UNIT DESIGN LABS

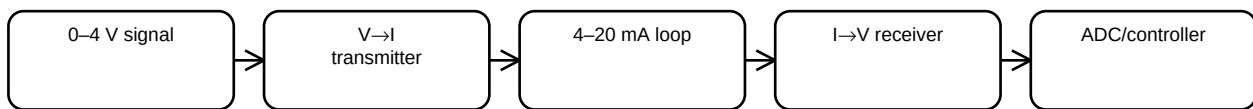
Lab 1 — Sensor to Display



Student workflow: identify sensor range → determine signal type → select conditioning → calculate gain/offset → select ADC/reference → scale digital code → display result → test endpoints.

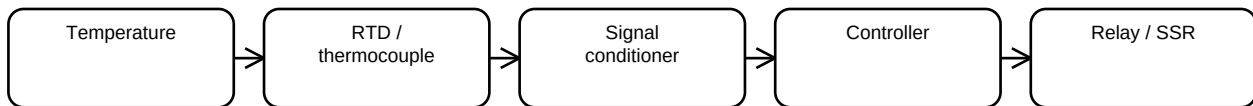
Fault injection: wrong gain, saturation, wrong reference, offset error, noisy signal, wrong display scaling.

Lab 2 — Industrial 4–20 mA Chain



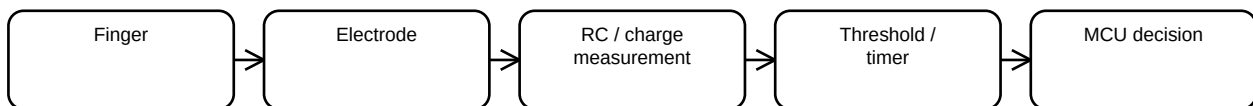
Make the learner calibrate 4 mA and 20 mA endpoints, then test intermediate points. Include broken-loop, wrong-burden and offset-error scenarios.

Lab 3 — Temperature Controller



Combine sensor conditioning, error amplifier, controller and actuator. Start with the supplied 0–400°C ON–OFF design, then let the student compare P/Pi/PID behavior.

Lab 4 — Touch Interface



Switch between self and mutual capacitance. Let the learner observe the physical quantity being measured and the digital decision process.

DEBUGGING LAB

Fault	What student should diagnose
Wrong op-amp resistor	Gain/output error; compare required ratio.
Excessive output request	Saturation; compare output with supply/output-swing limits.
Uncompensated VOS	Input-referred offset multiplied by noise gain.
Floating CMOS input	Undefined/noisy logic state.
TTL→CMOS level mismatch	Compare TTL VOH(min) with CMOS VIH(min).
No relay flyback protection	Inductive voltage reversal/spike risk.
Too-small ON–OFF hysteresis	Rapid relay switching near threshold.
ADC without adequate filtering	Aliasing from out-of-band components.
Wrong seven-segment polarity	Common-anode/common-cathode drive convention mismatch.
Excessive Cbase	Small relative touch-induced capacitance change.

Every debugging problem should provide progressive hints rather than immediate answers: Hint 1 → Hint 2 → Hint 3 → complete diagnosis.

EXAM MODE

2-mark format: definition + one key statement/equation.

4-mark format: definition/principle → labeled diagram → working → key equation/point.

8-mark/design format: requirements → assumptions → circuit/block diagram → derivation → component selection → calculation → verification → applications/limitations.

Priority design problems: 4–20 mA V–I; ON–OFF controller; K-type thermocouple conditioner; instrumentation amplifier; static/dynamic display; petrol measurement; ADC/reference/filter; capacitive touch.

Numerical interface: make the learner enter each intermediate result. When wrong, identify the failed step instead of only marking the final answer wrong.

VIVA QUESTION SET

- Why is the lower endpoint of a 4–20 mA signal not zero current?
- Why can a small op-amp offset become a large output error?
- Why is CMRR important for bridge/sensor measurements?
- Why should CMOS inputs not float?
- Why can TTL-to-CMOS require a pull-up?
- Why is hysteresis used in ON–OFF control?
- Why use an instrumentation amplifier for a bridge sensor?
- Why must anti-aliasing filtering occur before the ADC?
- Why does increasing ADC resolution reduce code-step size?
- Why does dynamic display rapidly switch digits?
- Why does a finger change sensor capacitance?
- Why can excessive baseline capacitance reduce touch sensitivity?
- Why does mutual capacitance change when a finger is between Tx and Rx?

ARENA AI IMPLEMENTATION BLUEPRINT

Reusable components to generate: ConceptCard, WhyButton, InteractiveCircuit, ParameterSlider, StepBuilder, NumericalSolver, WaveformGraph, DesignChallenge, DebugCircuit, QuizEngine and ProgressTracker.

Component	Required behavior
ConceptCard	Definition, intuition, formula, example and why.
WhyButton	Requirement → equation → selected value → what if changed.
InteractiveCircuit	Clickable vector/SVG circuit with live values.
StepBuilder	Reveals circuit/system one stage at a time.
NumericalSolver	Given → required → concept → equation → substitution → result → check.
WaveformGraph	Live time/frequency plot with annotations.
DesignChallenge	Requirement → student design → validation → hints → solution.
DebugCircuit	Fault → diagnosis → hints → explanation.
QuizEngine	MCQ, numerical, prediction, ordering, multi-select and debugging.
ProgressTracker	Stores completion, quiz results and weak areas locally.

Website lesson order: **Problem** → **Concept** → **Why** → **Diagram** → **Interaction** → **Calculation** → **What-if** → **Design** → **Quiz** → **Exam**. Do not create decorative controls that do nothing.

CONTENT INTEGRITY & SOURCE HANDLING

- Use the supplied course material as the primary source for course-specific values, terminology, examples and problems.
- Do not invent teacher-specific numerical values when the source does not provide them.
- Label additional explanation as Engineering Insight.
- Keep units visible in every calculation.
- When the source uses an approximation, label it as an approximation and preserve the source's stated comparison.
- Do not silently replace the source's terminology or action/sign convention.
- Use source diagrams as the basis for interactive reconstructions, but label a redrawn circuit as an interactive reconstruction.
- Do not claim a generated diagram or numerical value came from the source unless it is actually supported by the source.

SOURCE-DERIVED PROBLEM INDEX

Unit	Major supplied problems/design tasks
1	Reliability/exponential law; bathtub curve; System Performance Matrix and Design Matrix; inverting amplifier practical current/power;
2	4–20 mA V–I; pneumatic ON–OFF controller; K-type thermocouple conditioner; instrumentation-amplifier gain; load-cell specifications
3	Static seven-segment; dynamic seven-segment; petrol/diesel volume measurement; ADC/DAC specifications; 10-bit 4.094 V ADC; g

FINAL WEBSITE REQUIREMENTS

- Multi-page responsive application; not a single long notes page.
- All syllabus topics represented.
- PDF/PPT-derived material integrated when provided; syllabus alone remains sufficient to run the course.
- Interactive circuits wherever practical.
- Live component/value changes with calculations and visual effects.
- Step-by-step derivations rather than formula dumping.
- Design problems require student interaction before revealing the full solution.
- Debugging mode with intentional faults.
- Quiz and exam mode.
- Local progress tracking.
- Clear distinction between Course Material and Engineering Insight.
- No fake buttons or placeholder lorem ipsum.
- Prefer a working simpler simulation over a visually impressive non-functional one.

The student should repeatedly be able to ask: **Why this component? Why this topology? Why this value? Why this reference? Why this gain? Why this filter? What happens if I change it?** The interactive website exists to answer those questions visually, mathematically and intuitively.

END OF ESD MASTER KNOWLEDGE DOCUMENT

Primary source: TY E&TC; Electronic system design (PE), Course code UETPE0513, "TY ESD 2026 ISE1 (20 files merged).pdf", supplied in this conversation. The source is a 205-page merged PDF containing 2026 assignment/course material plus dated lecture/slide material from earlier years.